

Unrepresentative big surveys significantly overestimated US vaccine uptake

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 Check for updates

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Surveys are a crucial tool for understanding public opinion and behaviour, and their accuracy depends on maintaining statistical representativeness of their target populations by minimizing biases from all sources. Increasing data size shrinks confidence intervals but magnifies the effect of survey bias: an instance of the Big Data Paradox¹. Here we demonstrate this paradox in estimates of first-dose COVID-19 vaccine uptake in US adults from 9 January to 19 May 2021 from two large surveys: Delphi–Facebook^{2,3} (about 250,000 responses per week) and Census Household Pulse⁴ (about 75,000 every two weeks). In May 2021, Delphi–Facebook overestimated uptake by 17 percentage points (14–20 percentage points with 5% benchmark imprecision) and Census Household Pulse by 14 (11–17 percentage points with 5% benchmark imprecision), compared to a retroactively updated benchmark the Centers for Disease Control and Prevention published on 26 May 2021. Moreover, their large sample sizes led to miniscule margins of error on the incorrect estimates. By contrast, an Axios–Ipsos online panel⁵ with about 1,000 responses per week following survey research best practices⁶ provided reliable estimates and uncertainty quantification. We decompose observed error using a recent analytic framework¹ to explain the inaccuracy in the three surveys. We then analyse the implications for vaccine hesitancy and willingness. We show how a survey of 250,000 respondents can produce an estimate of the population mean that is no more accurate than an estimate from a simple random sample of size 10. Our central message is that data quality matters more than data quantity, and that compensating the former with the latter is a mathematically provable losing proposition.

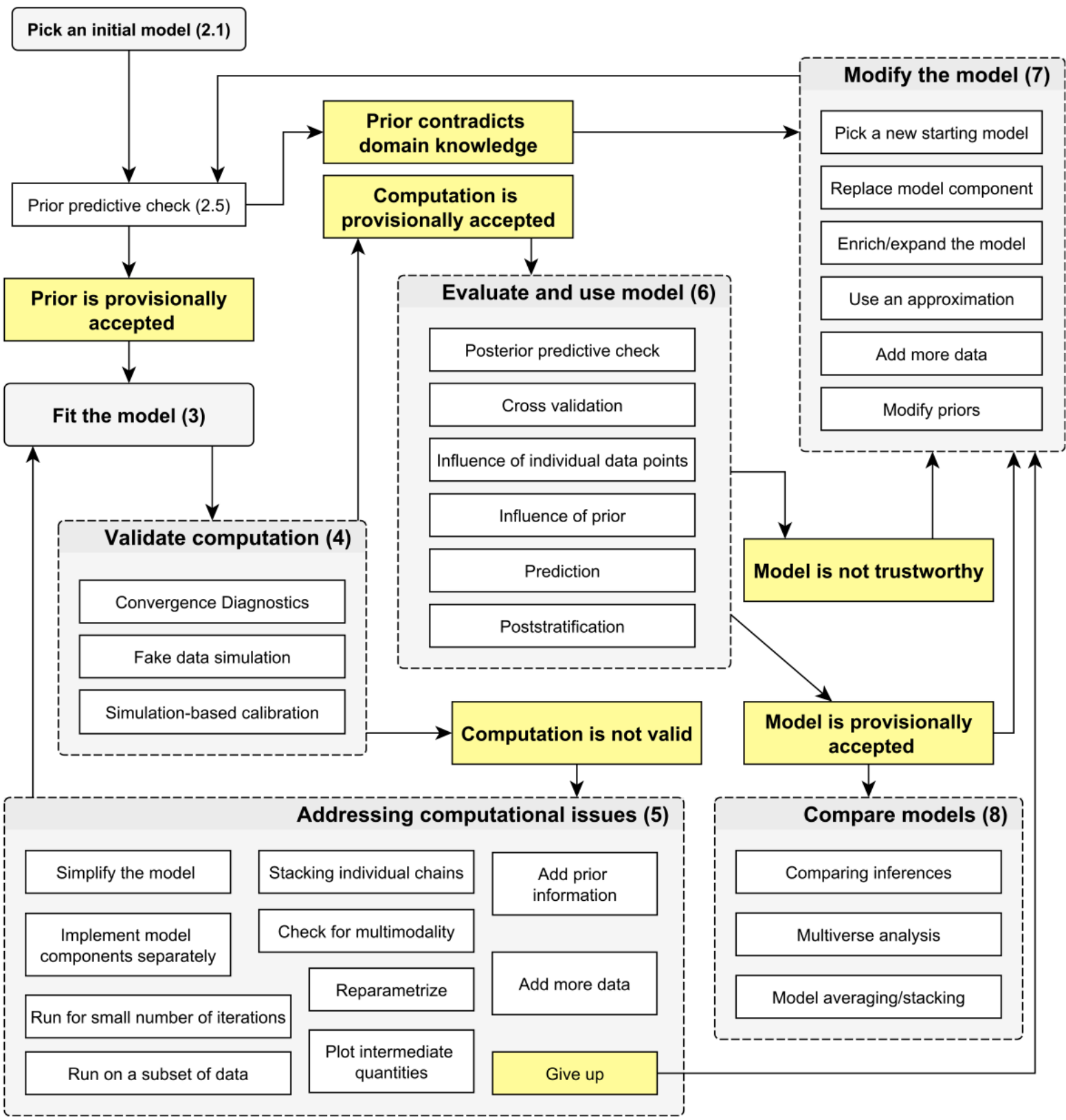
Governments, businesses and researchers rely on survey data to inform the provision of government services⁷, steer business strategy and guide the response to the COVID-19 pandemic^{8,9}. With the ever-increasing volume and accessibility of online surveys and organically collected data, the line between traditional survey research and Big Data is becoming increasingly blurred¹⁰. Large datasets enable the analysis of fine-grained subgroups, which are in high demand for designing targeted policy interventions¹¹. However, counter to common intuition¹², larger sample sizes alone do not ensure lower error. Instead, small biases are compounded as sample size increases¹.

We see initial evidence of this in the discrepancies in estimates of first-dose COVID-19 vaccine uptake, willingness and hesitancy from three online surveys in the US. Two of them—Delphi–Facebook’s COVID-19 symptom tracker^{2,3} (around 250,000 responses per week and with over 4.5 million responses from January to May 2021) and the Census Bureau’s Household Pulse survey⁴ (around 75,000 responses per survey wave and with over 600,000 responses from January to May 2021)—have large enough sample sizes to render standard uncertainty intervals negligible; however, they report significantly different estimates of vaccination

behaviour with nearly identically worded questions (Table 1). For example, Delphi–Facebook’s state-level estimates for willingness to receive a vaccine from the end of March 2021 are 8.5 percentage points lower on average than those from the Census Household Pulse (Extended Data Fig. 1a), with differences as large as 16 percentage points.

The US Centers for Disease Control and Prevention (CDC) compiles and reports vaccine uptake statistics from state and local offices¹³. These figures serve as a rare external benchmark, permitting us to compare survey estimates of vaccine uptake to those from the CDC. The CDC has noted the discrepancies between their own reported vaccine uptake and that of the Census Household Pulse^{14,15}, and we find even larger discrepancies between the CDC and Delphi–Facebook data (Fig. 1a). By contrast, the Axios–Ipsos Coronavirus Tracker⁵ (around 1,000 responses per wave, and over 10,000 responses from January to May 2021) tracks the CDC benchmark well. None of these surveys use the CDC benchmark to adjust or assess their estimates of vaccine uptake, thus by examining patterns in these discrepancies, we can infer each survey’s accuracy and statistical representativeness, a nuanced concept that is critical for the reliability of survey findings^{16–19}.

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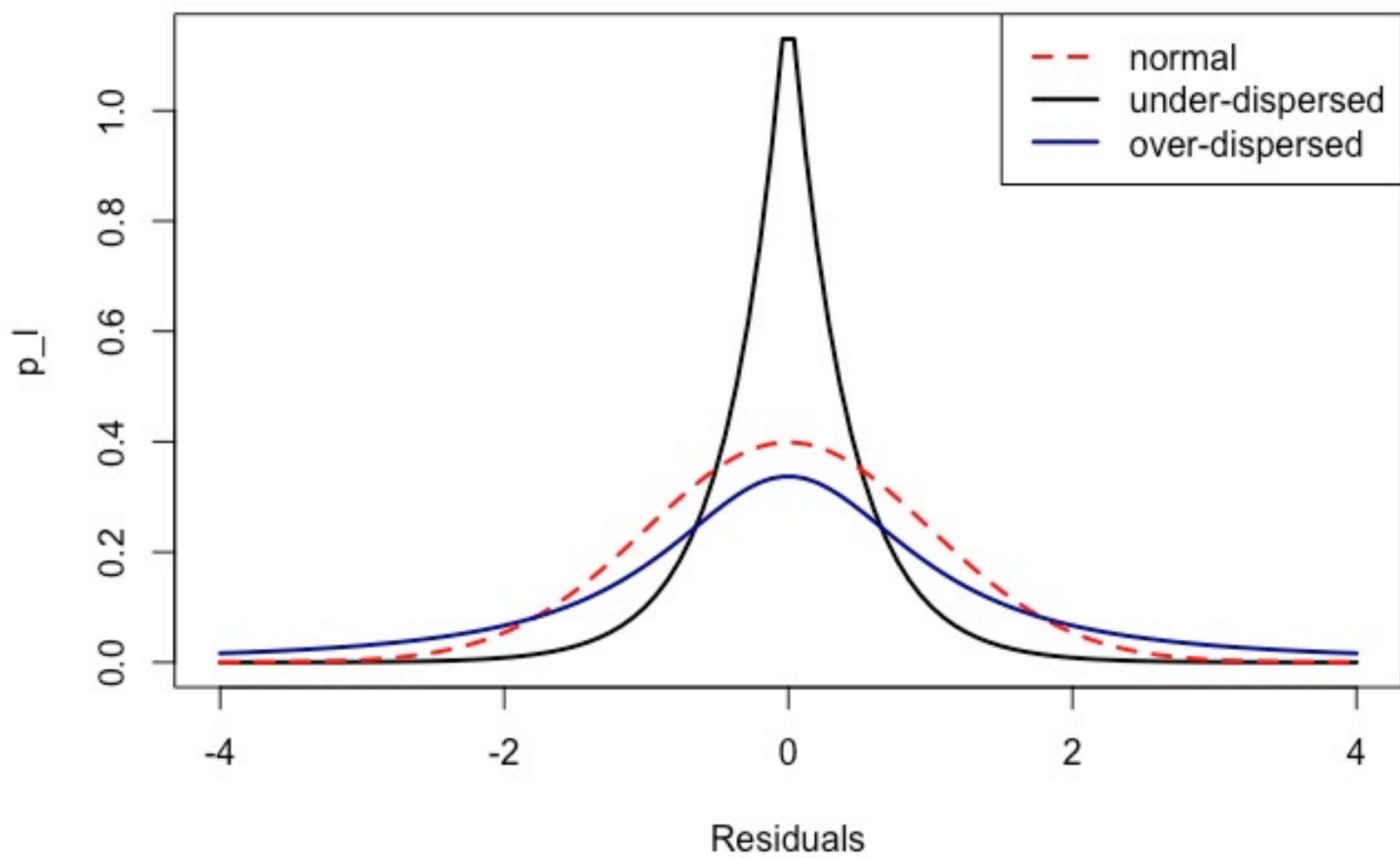
Monsters and mixtures

- More complicated GLMs:
 - *Monsters*: Specialized, complex distributions
 - ordered categories, ranks
 - *Mixtures*: Blends of stochastic processes
 - Varying means, probabilities, rates
 - Varying process: zero-inflation, hurdles

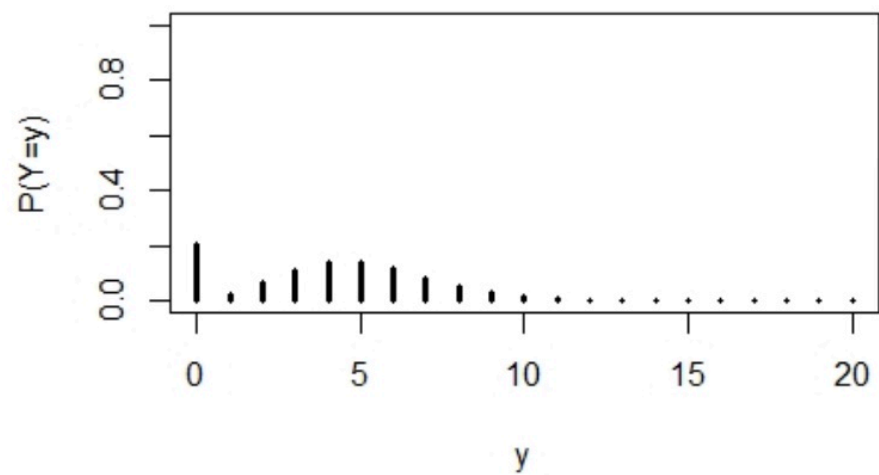


Mixtures

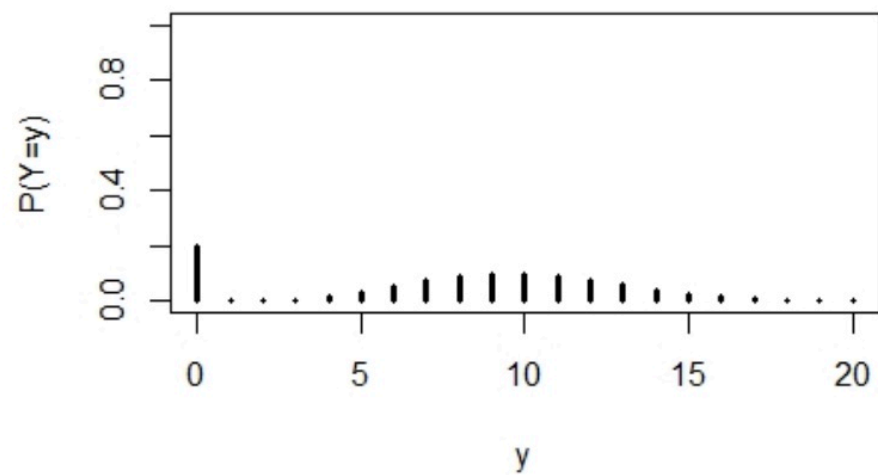
- Some outcomes mix different processes
 - replace parameter of likelihood with distribution of its own
- Over-dispersion: counts often more variable than expected, because probabilities/rates are variable
 - beta-binomial, gamma-Poisson (negative-binomial)
- Zero-inflated mixtures



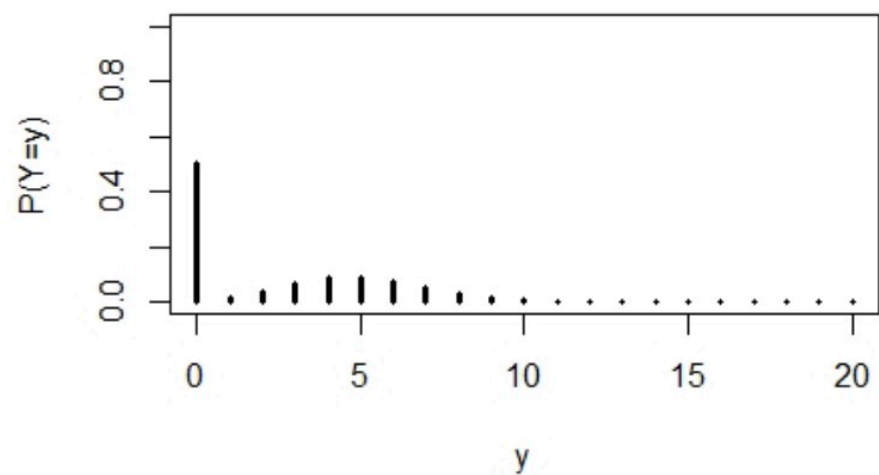
$\pi=0.2, \mu=5$



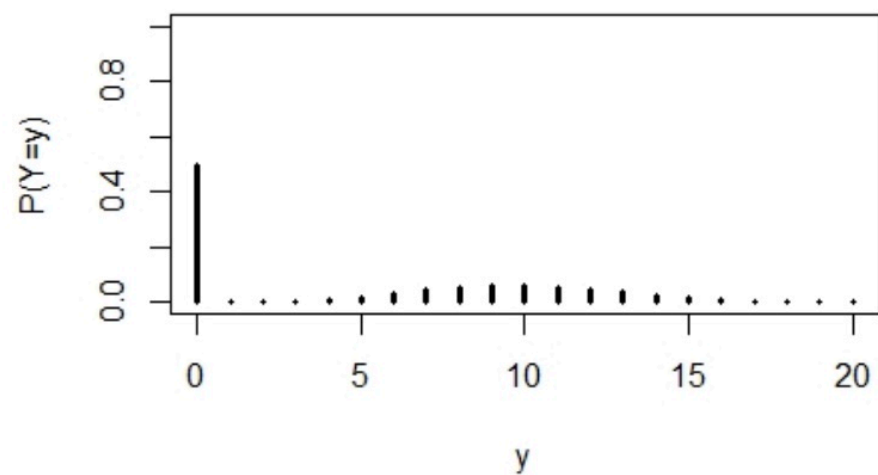
$\pi=0.2, \mu=10$



$\pi=0.5, \mu=5$



$\pi=0.5, \mu=10$



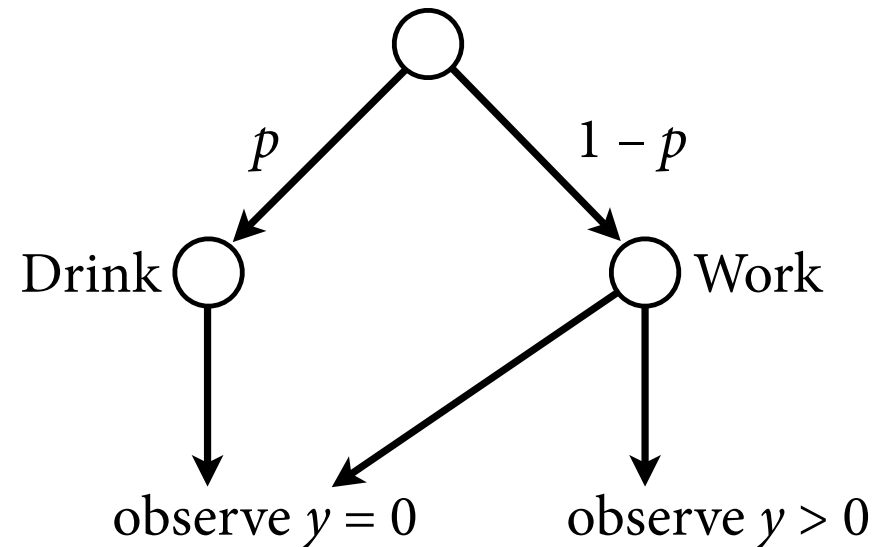
Zero-inflated Poisson model

$$y_i \sim \text{ZIPoisson}(p_i, \lambda_i)$$

$$\text{logit}(p_i) = \alpha_p + \beta_p x_i$$

$$\log(\lambda_i) = \alpha_\lambda + \beta_\lambda x_i$$

*Linear models
are independent*



Simulate, validate, cromulate

- As models get more complicated, no guarantees you can
 - specify model correctly
 - estimate actual process reliably
 - Bayes not magic, just logic
- Simulate “dummy data”
 - recover estimates
 - understand the model
- Try parameter combinations hostile to estimation, so you know limits of the golem



Other mixtures

- Can ZIBinomial, too
- Also “hurdle” models, aka zero-augmented
- Continuous mixtures for overdispersed counts
 - beta-binomial
 - gamma-Poisson
- Multilevel models are a sort of mixture

```
zero_inflated_beta(link = "logit", link_phi = "log", link_zi = "logit")
```

```
zero_one_inflated_beta(  
  link = "logit",  
  link_phi = "log",  
  link_zoi = "logit",  
  link_coi = "logit"  
)
```

```
zero_inflated_poisson(link = "log", link_zi = "logit")
```

```
zero_inflated_negbinomial(link = "log", link_shape = "log", link_zi = "logit")
```

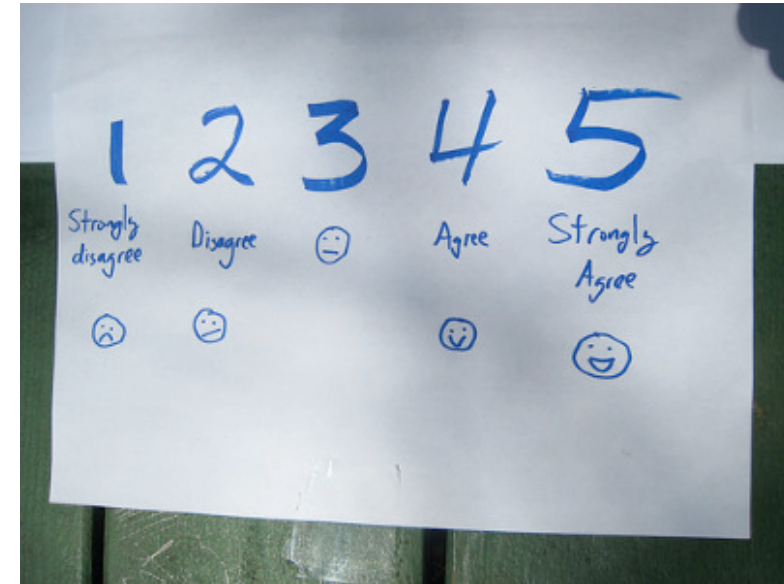
```
zero_inflated_binomial(link = "logit", link_zi = "logit")
```

```
zero_inflated_beta_binomial(  
  link = "logit",  
  link_phi = "log",  
  link_zi = "logit"  
)
```

Ordered categories

- How much do you like this class? (1–7)
- How important is income of a potential spouse? (1–10)
- How often do you see bats? (never, sometimes, frequently)
- Depth harbor seals dive? (shallow, middle, deep)

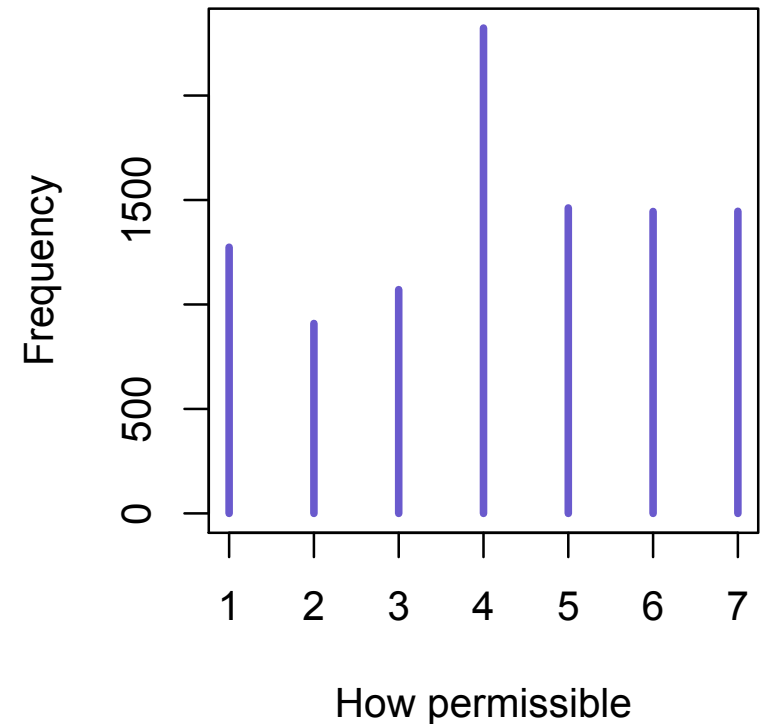
Ordered categories



- Discrete outcomes
- Defined minimum and maximum
- Defined order
- “Distances” between categories unknown

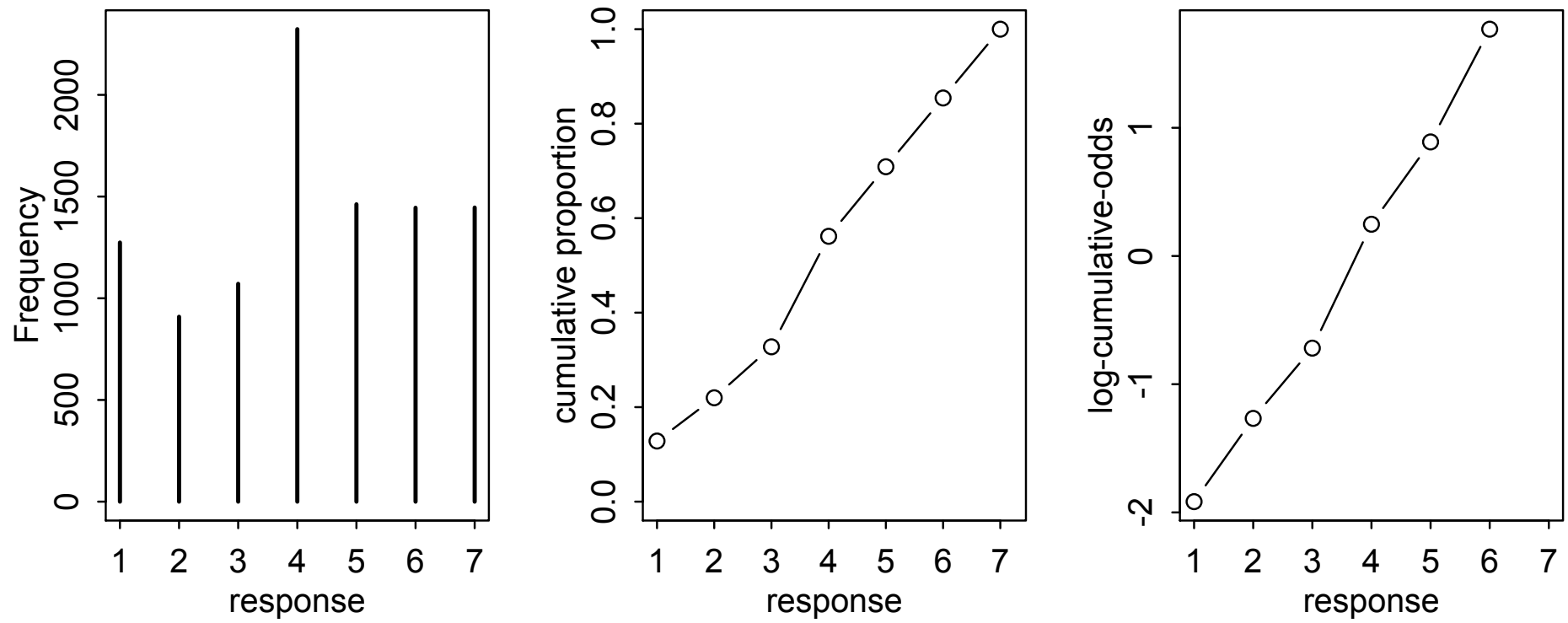
Moral intuitions

- data(Trolley)
- 331 individuals, 30 scenarios, 9930 responses
- How do responses vary with action, intention, contact?
- Age, gender, individual?



Ordered logit

- A log-cumulative-odds link probability model



Ordered logit

- A log-cumulative-odds link probability model

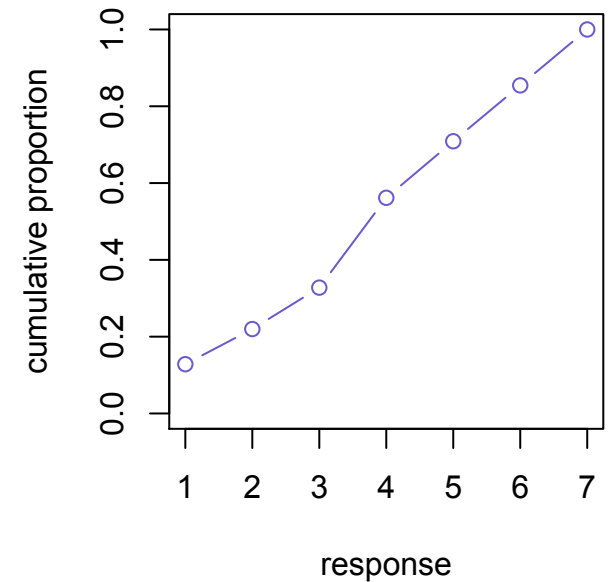
cumulative log-odds

response

category

$$\log \frac{\Pr(y_i \leq k)}{1 - \Pr(y_i \leq k)} = \phi_k$$

linear model



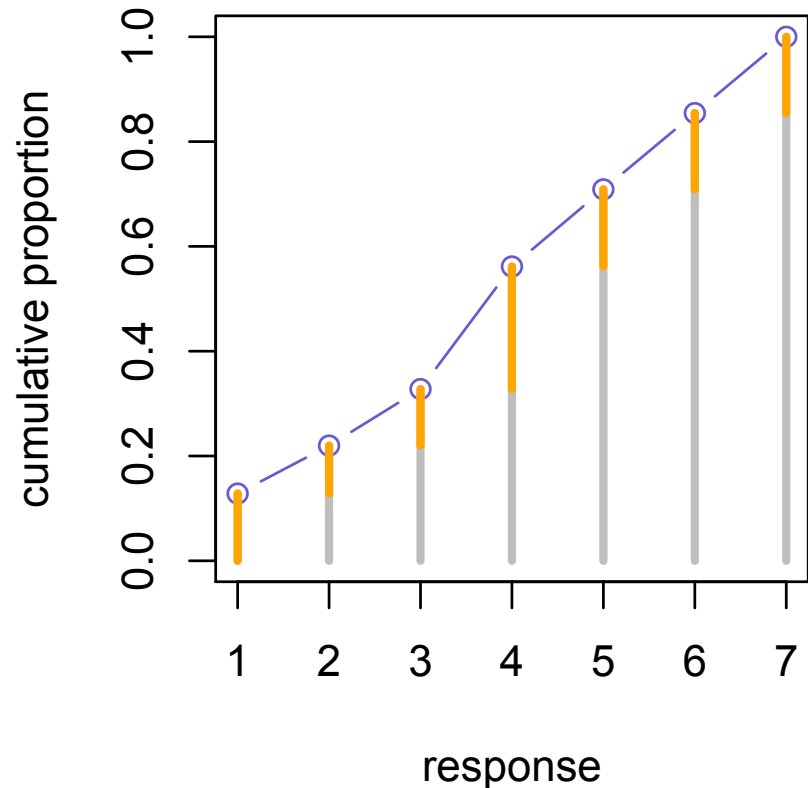
Ordered logit

- A log-cumulative-odds link probability model

$$\log \frac{\Pr(y_i \leq k)}{1 - \Pr(y_i \leq k)} = \phi_k$$

$$\Pr(y_i \leq k) = \frac{\exp(\phi_k)}{1 + \exp(\phi_k)}$$

$$\Pr(y_i = k) = \Pr(y_i \leq k) - \Pr(y_i \leq k - 1)$$



Ordered logit

```
m12.5 <- ulam(  
  alist(  
    R ~ dordlogit( 0 , cutpoints ),  
    cutpoints ~ dnorm( 0 , 1.5 )  
  ) ,  
  data=list( R=d$response ), chains=4 , cores=3 )
```

$R_i \sim \text{Categorical}(\mathbf{p})$ [probability of data]

$p_1 = q_1$ [probabilities of each value k]

$p_k = q_k - q_{k-1}$ for $K > k > 1$

$p_K = 1 - q_{K-1}$

$\text{logit}(q_k) = \kappa_k - \phi_i$ [cumulative logit link]

$\phi_i = \text{terms of linear model}$ [linear model]

$\kappa_k \sim \text{Normal}(0, 1.5)$ [common prior for each intercept]

Back to probability scale

```
precis( m12.5 , depth=2 )
```

R code
12.18

	mean	sd	5.5%	94.5%	n_eff	Rhat
cutpoints[1]	-1.92	0.03	-1.96	-1.87	1460	1
cutpoints[2]	-1.27	0.02	-1.31	-1.23	2091	1
cutpoints[3]	-0.72	0.02	-0.75	-0.68	2480	1
cutpoints[4]	0.25	0.02	0.22	0.28	2701	1
cutpoints[5]	0.89	0.02	0.85	0.92	2373	1
cutpoints[6]	1.77	0.03	1.72	1.81	2345	1

```
inv_logit(coef(m12.5))
```

R code
12.19

cutpoints[1]	cutpoints[2]	cutpoints[3]	cutpoints[4]	cutpoints[5]	cutpoints[6]
0.1283325	0.2198127	0.3276819	0.5615751	0.7087310	0.8543559

Adding predictor variables

$$\log \frac{\Pr(y_i \leq k)}{1 - \Pr(y_i \leq k)} = \alpha_k - \phi_i$$

$$\phi_i = \beta_A A_i + \beta_C C_i + B_{I,i} I_i$$

$$B_{I,i} = \beta_I + \beta_{IA} A_i + \beta_{IC} C_i$$

```
m12.6 <- ulam(  
  alist(  
    R ~ dordlogit( phi , cutpoints ),  
    phi <- bA*A + bC*C + BI*I ,  
    BI <- bI + bIA*A + bIC*C ,  
    c(bA,bI,bC,bIA,bIC) ~ dnorm( 0 , 0.5 ),  
    cutpoints ~ dnorm( 0 , 1.5 )  
  ) , data=dat , chains=4 , cores=4 )
```

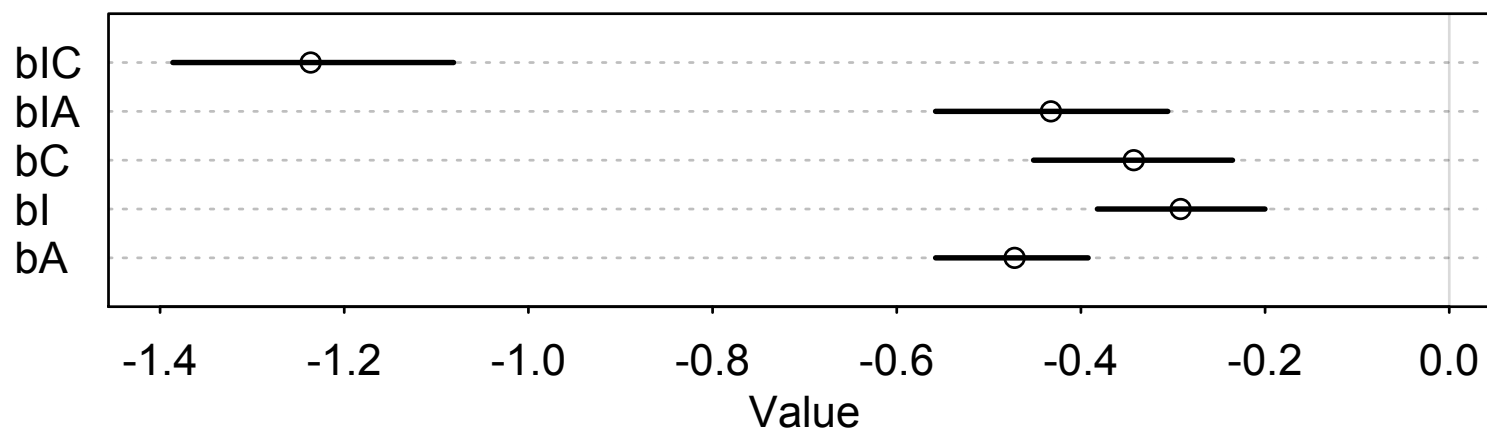
```

m12.6 <- ulam(
  alist(
    R ~ dordlogit( phi , cutpoints ),
    phi <- bA*A + bC*C + BI*I ,
    BI <- bI + bIA*A + bIC*C ,
    c(bA,bI,bC,bIA,bIC) ~ dnorm( 0 , 0.5 ),
    cutpoints ~ dnorm( 0 , 1.5 )
  ) , data=dat , chains=4 , cores=4 )
precis( m12.6 )

```

6 vector or matrix parameters omitted in display. Use depth=2 to show them.

	mean	sd	5.5%	94.5%	n_eff	Rhat
bIC	-1.24	0.09	-1.39	-1.08	1038	1
bIA	-0.43	0.08	-0.56	-0.31	927	1
bC	-0.34	0.07	-0.45	-0.24	1168	1
bI	-0.29	0.06	-0.38	-0.20	806	1
bA	-0.47	0.05	-0.56	-0.39	1042	1



Plotting ordered logits

- Oh, bother: Posterior prediction a *vector* of probabilities, one for each level of outcome
- How to plot this?
- Several useful options



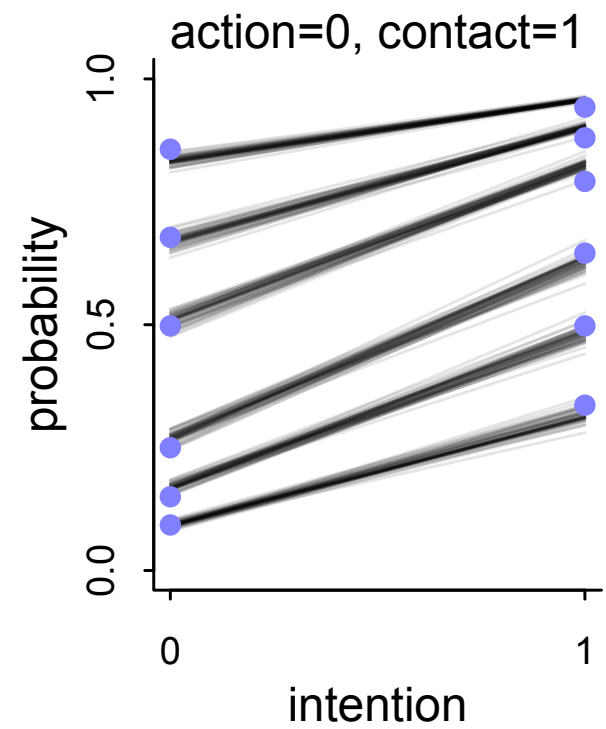
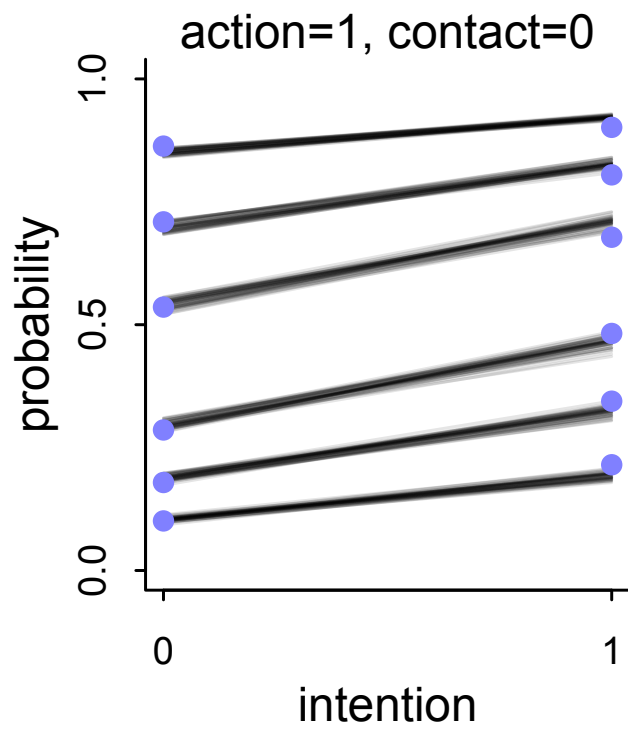
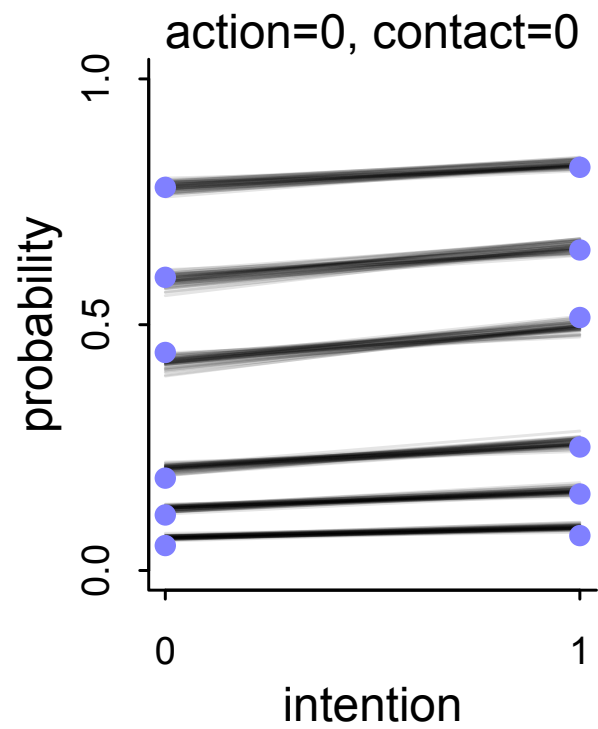


Figure 12.6

Ordered categorical predictors

- Not just outcomes, also predictors
- Ordinary metric predictor assumes each unit change same change (on linear scale)
- Ordered category predictor: Each level could have different size of influence, but all effects in same direction (monotonic)
 - Example: Education

```
library(rethinking)
data(Trolley)
d <- Trolley
levels(d$edu)
```

R code
12.30

```
[1] "Bachelor's Degree"    "Elementary School"    "Graduate Degree"
[4] "High School Graduate" "Master's Degree"      "Middle School"
[7] "Some College"        "Some High School"
```

Ordered categorical predictors

```
library(rethinking)
data(Trolley)
d <- Trolley
levels(d$edu)
```

R code
12.30

```
[1] "Bachelor's Degree"      "Elementary School"    "Graduate Degree"
[4] "High School Graduate"  "Master's Degree"      "Middle School"
[7] "Some College"          "Some High School"
```

- Strategy
 - Each level gets a unique parameter
 - Parameters sum to total maximum effect of education
 - Assign prior to total effect and to *proportions* of total effect

Ordered categorical predictors

- Make all deltas sum to 1, and use leading coefficient for maximum effect:

$$\phi_i = \underbrace{\beta_E}_{\substack{\text{maximum effect} \\ \text{of education}}} \sum_{j=0}^{E_i-1} \delta_j + \text{other stuff}$$

Ordered categorical predictors

- Make all deltas sum to 1, and use leading coefficient for maximum effect:

$$R_i \sim \text{Ordered-logit}(\phi_i, \kappa)$$

$$\phi_i = \beta_E \sum_{j=0}^{E_i-1} \delta_j + \beta_A A_i + \beta_I I_i + \beta_C C_i$$

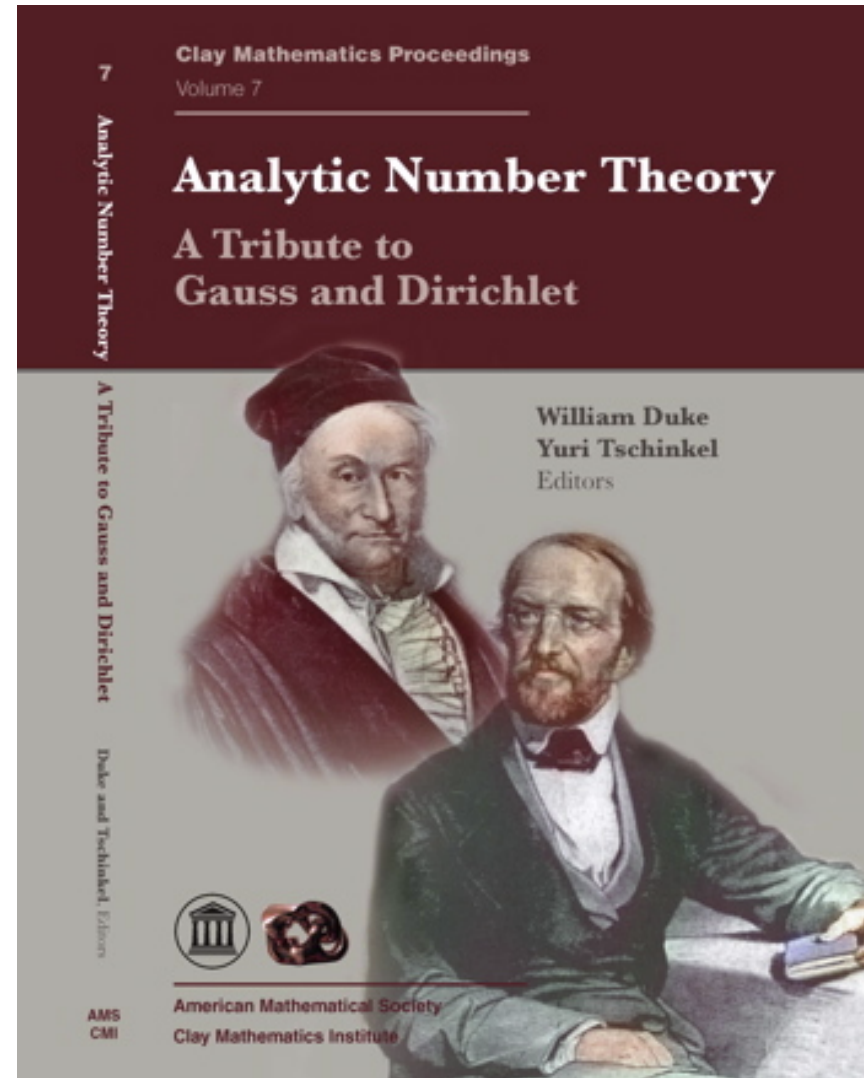
$$\kappa_k \sim \text{Normal}(0, 1.5)$$

$$\beta_A, \beta_I, \beta_C, \beta_E \sim \text{Normal}(0, 1)$$

$$\delta \sim \text{Dirichlet}(\alpha)$$

Dirichlet (dee-ree-klay)

- Dirichlet: Distribution of N probabilities
- A distribution of distributions
- Generalization of beta distribution
- Shape determined by vector of N parameters
- Each parameter is a *pseudo-count*
 - Large value means that category more probable



Johann Peter Gustav **Lejeune Dirichlet**
(1805–1859)